



AMAZON
WEB SERVICES

MACHINE LEARNING FOR ATTACK CLASSIFICATION

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OUR TEAM



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BUSINESS USE CASE



- \$200 billion industry
 - AI: 23.5% CAGR
- Main Targets
 - Threats with Increasing Sophistication
 - Inadequate Security Practices
 - Rapid Technological Changes

PROJECT SCOPE



- Model Adaptation
 - New Threats
- Scalability
 - Handling Larger Inputs
 - Real-time Processing
- Intrusion Detection Systems (IDS)
 - Security Solutions
 - Risk Mitigation

AWS TOOLS



**Amazon
SageMaker**



**amazon
S3**

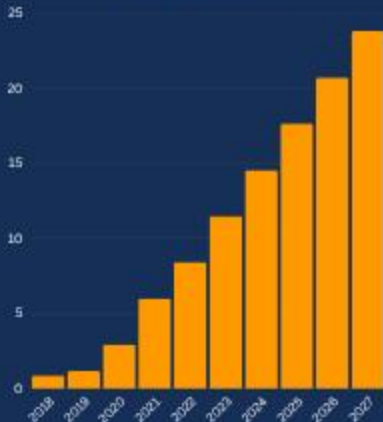
WHY OUR DATASET

Cybersecurity is prevalent today with the wide spread usage of technology and AI.

Crucial data for our model:

- What kind of packets are attacked
- How long it took to get into certain packets

Estimated cost of cybercrime worldwide
(in trillion US dollars)



Source: World Economic Forum

OUR DATASET

- Courtesy of **UC Irvine Archives**
- **123,117** samples
- **12** classes
- **83** features
- Collected data from various IoT devices
- **Collection:** Zeek monitoring tool.
- Numerical and String type data

file_tot	file_avg	file	init_windex	init_windex	z_wi	Attack_type
28729182.56	28729182.96	0	64240	26847	502 MQTT_Publish	
28855277.06	28855277.06	0	64240	26847	502 MQTT_Publish	
28842149.02	28842149.02	0	64240	26847	502 MQTT_Publish	
28913774.97	28913774.97	0	64240	26847	502 MQTT_Publish	
28814704.9	28814704.9	0	64240	26847	502 MQTT_Publish	
28825548.17	28825548.17	0	64240	26847	502 MQTT_Publish	
28823416.95	28823416.95	0	64240	26847	502 MQTT_Publish	
28795433.04	28795433.04	0	64240	26847	502 MQTT_Publish	
28661617.04	28661617.04	0	64240	26847	502 MQTT_Publish	
28778450.97	28778450.97	0	64240	26847	502 MQTT_Publish	
28870183.54	28870183.54	0	64240	26847	502 MQTT_Publish	
28819482.09	28819482.09	0	64240	26847	502 MQTT_Publish	
28747197.87	28747197.87	0	64240	26847	502 MQTT_Publish	
28850662.95	28850662.95	0	64240	26847	502 MQTT_Publish	
28860140.09	28860140.09	0	64240	26847	502 MQTT_Publish	
28770004.03	28770004.03	0	64240	26847	502 MQTT_Publish	
28775004.86	28775004.86	0	64240	26847	502 MQTT_Publish	
28875375.08	28875375.08	0	64240	26847	502 MQTT_Publish	
28776599.17	28776599.17	0	64240	26847	502 MQTT_Publish	
28892142.06	28892142.06	0	64240	26847	502 MQTT_Publish	
28772324.09	28772324.09	0	64240	26847	502 MQTT_Publish	
28765208.96	28765208.96	0	64240	26847	502 MQTT_Publish	
28867868.9	28867868.9	0	64240	26847	502 MQTT_Publish	
28697514.06	28697514.06	0	64240	26847	502 MQTT_Publish	
28937090.16	28937090.16	0	64240	26847	502 MQTT_Publish	
28873291.02	28873291.02	0	64240	26847	502 MQTT_Publish	
28746304	28746304	0	64240	26847	502 MQTT_Publish	
28878719.09	28878719.09	0	64240	26847	502 MQTT_Publish	
28794428.11	28794428.11	0	64240	26847	502 MQTT_Publish	
28802947.04	28802947.04	0	64240	26847	502 MQTT_Publish	
28799875.97	28799875.97	0	64240	26847	502 MQTT_Publish	
28783406.97	28783406.97	0	64240	26847	502 MQTT_Publish	
28791519.17	28791519.17	0	64240	26847	502 MQTT_Publish	
28870038.03	28870038.03	0	64240	26847	502 MQTT_Publish	

PRE PROCESSING



Data Encoding

Split Data into Flavors

Original Data

Normalized Data

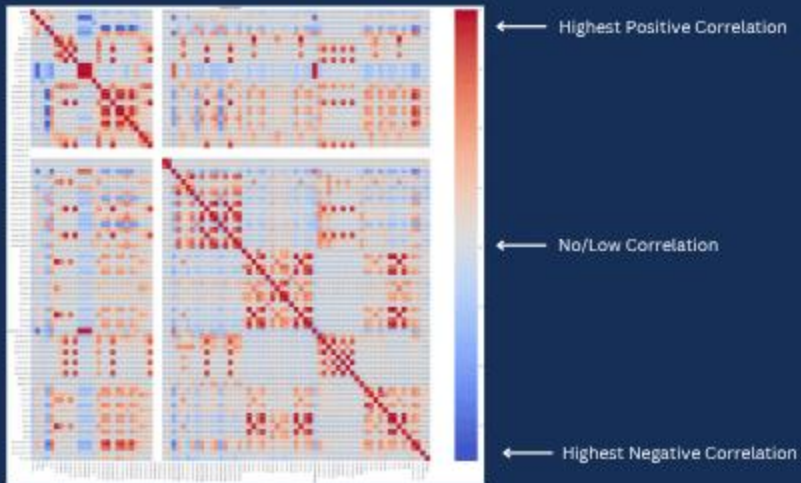
Highly different range of
numbers

Resampled Data

Highly unbalanced
data

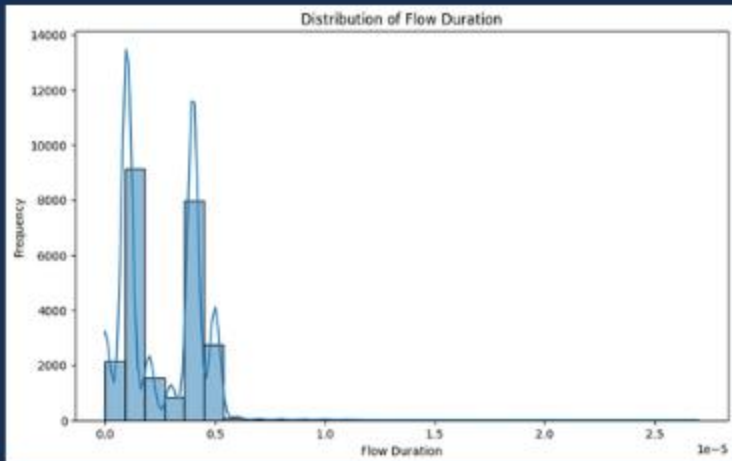
DATA VISUALIZATION (EDA)

01 Heatmap of Dataset



DATA VISUALIZATION (EDA)

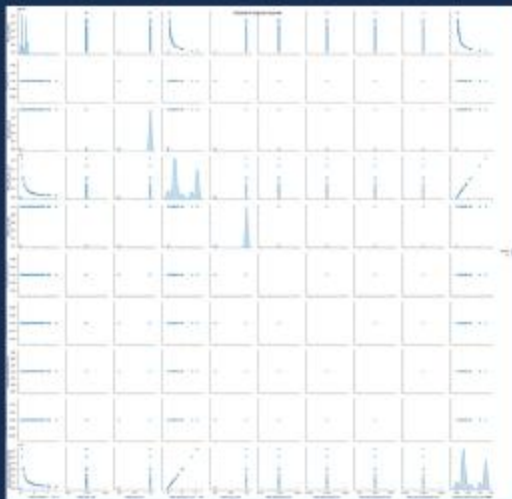
02 Distribution of Flow Duration



DATA VISUALIZATION (EDA)

03

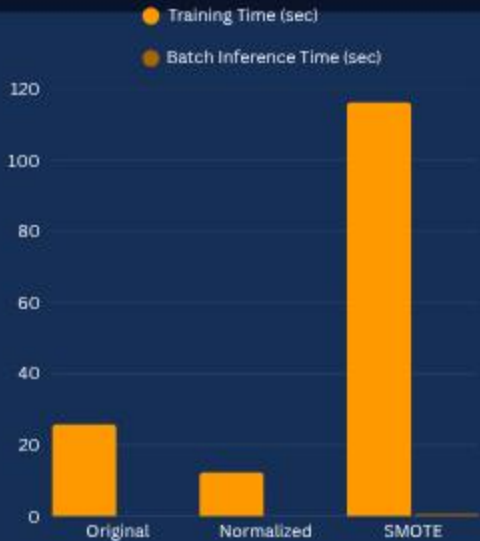
Pair Plot of RT-
IoT2022 Dataset



ALGORITHMS

01

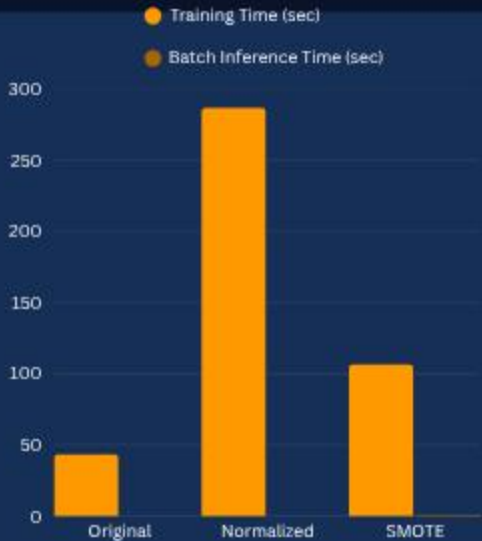
Logistic Regression Algorithm



ALGORITHMS

02

Neural Networks Algorithm



ALGORITHMS

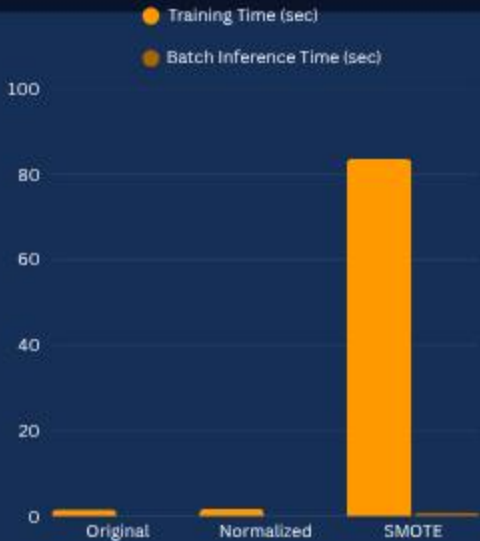
03 KNN



*SMOTE did not load in a reasonable amount of time

ALGORITHMS

04 Decision Tree Algorithm



ALGORITHMS

03 Decision Tree Algorithm



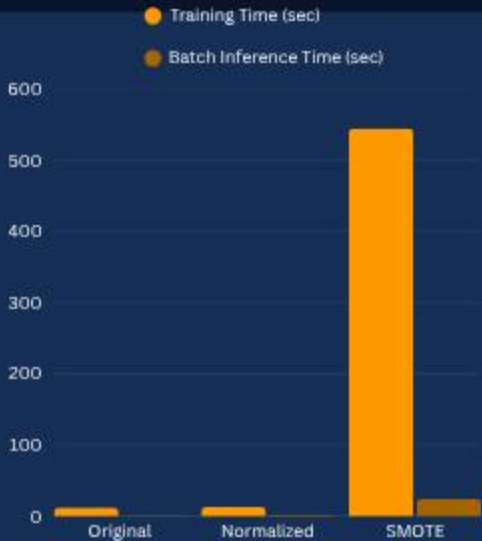
Original Decision Tree Algorithm



Decision Tree Feature Importance

ALGORITHMS

05 Random Forest Classifier Algorithm



RESULTS AND IMPACT



- Extensive data pre-processing and debugging -->
 - accuracy rates b/w 98% - 100%
- Decrease attacks for:
 - consumers
 - businesses



THANK YOU

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